

SUTHERLAND, South Africa

WMO: 687220

Lat: 32.400S

Lon: 20.663E

Elev: 4783

StdP: 12.33

Time Zone: 2.00 (E02)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.4	24.2	7.7	9.7	57.9	11.0	11.5	54.2	22.0	48.8	20.1	53.8	1.0	210	0.477

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.7	88.9	57.4	86.2	56.4	83.8	55.8	62.1	75.0	61.0	74.7	59.8	74.3	8.3	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
58.4	87.3	66.0	56.6	81.7	65.9	54.5	75.6	65.1	30.4	75.1	29.4	74.5	28.5	74.1	67.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.0	16.6	14.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	55.4	68.5	67.5	63.4	56.1	49.7	43.5	42.9	44.1	49.9	55.4	59.7	65.2
	DBStd	10.81	5.06	5.37	5.85	6.40	5.59	5.32	6.42	5.57	5.55	7.10	6.75	5.47
	HDD50	841	0	0	2	18	76	207	236	199	69	26	7	1
	HDD65	3891	24	33	99	272	476	645	686	649	453	307	182	65
	CDD50	2821	574	489	417	202	65	12	16	16	66	195	298	471
	CDD65	396	133	102	49	6	0	0	0	0	0	11	24	71
	CDH74	5635	1639	1294	663	114	10	0	0	1	31	296	499	1088
	CDH80	1607	556	398	148	7	0	0	0	0	1	73	111	313
(14) Wind	WSAvg	6.1	6.3	5.9	5.5	5.4	5.8	6.8	6.3	5.9	6.1	6.5	6.5	6.2
(15) Precipitation	PrecAvg	10.4	0.6	0.7	1.3	1.1	0.9	1.4	1.0	0.9	0.7	0.6	0.7	0.6
	PrecMax	19.5	4.5	3.7	4.4	4.1	3.4	4.3	3.4	2.5	3.1	2.5	3.5	3.4
	PrecMin	5.2	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0
	PrecStd	3.1	0.8	0.8	1.1	1.1	0.9	1.1	0.8	0.7	0.6	0.6	0.8	0.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	92.6	91.6	87.9	80.8	75.2	65.6	67.6	73.2	78.0	88.1	87.9	90.0
		MCWB	58.4	58.7	57.8	55.1	51.5	46.6	46.9	49.0	50.6	55.8	55.0	57.2
	2%	DB	88.9	87.6	84.3	77.2	69.8	62.0	62.9	65.4	73.1	81.2	82.8	86.4
		MCWB	57.6	57.6	56.2	53.6	49.4	45.9	46.0	49.3	52.9	53.8	55.7	57.2
	5%	DB	85.8	84.8	81.2	73.5	65.1	58.0	59.5	61.2	68.7	77.2	79.6	83.6
		MCWB	56.3	56.9	55.5	52.8	47.9	44.2	44.1	44.2	46.8	51.4	52.8	55.0
	10%	DB	82.9	82.0	77.9	69.5	61.3	54.4	56.0	57.1	64.3	72.9	76.2	80.5
		MCWB	56.1	56.2	54.7	51.7	47.2	42.9	43.3	42.8	45.6	49.9	51.8	54.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.4	63.3	64.3	59.3	54.4	51.0	49.4	51.1	53.1	58.4	60.1	61.7
		MCDB	74.2	76.8	79.2	72.3	62.3	59.0	61.7	62.5	73.4	73.9	75.8	74.3
	2%	WB	61.9	61.7	61.8	57.4	52.5	48.1	47.3	48.4	50.3	56.4	56.7	60.1
		MCDB	74.7	76.6	72.8	69.4	61.8	55.3	59.0	63.4	68.0	75.0	74.2	73.8
	5%	WB	60.7	60.4	59.6	55.9	50.9	46.2	45.6	45.7	48.4	54.1	55.0	58.7
		MCDB	75.5	75.6	71.5	67.9	60.2	54.7	56.8	57.9	64.1	70.7	73.5	74.2
	10%	WB	59.5	59.1	57.7	53.9	49.2	44.5	44.1	43.5	46.6	52.0	53.3	57.3
		MCDB	75.3	74.7	71.9	65.5	58.7	52.9	53.8	55.3	61.8	68.7	71.7	74.4
(35) Mean Daily Temperature Range	5% DB	MDBR	26.7	27.7	27.8	25.8	23.0	20.5	22.3	24.7	27.6	28.6	27.6	27.6
		MCDBR	31.4	31.9	32.1	32.4	30.6	28.0	28.7	32.1	36.0	37.0	33.8	33.1
	5% WB	MCWBR	9.4	10.9	11.5	13.5	14.7	15.1	14.9	15.5	16.5	14.7	11.6	11.1
		MCWBR	23.7	26.0	24.4	24.9	22.6	22.2	24.9	27.3	31.2	30.2	28.7	24.6
(40) Clear-Sky Solar Irradiance	taub	taub	0.285	0.288	0.283	0.268	0.255	0.244	0.247	0.266	0.278	0.292	0.277	0.281
		taud	2.635	2.629	2.639	2.700	2.723	2.719	2.693	2.592	2.520	2.509	2.628	2.640
	Ebn at Noon	Ebn at Noon	335	328	319	309	298	295	298	304	315	323	335	337
		Edh at Noon	32	31	29	24	21	19	21	26	31	34	32	32
(44) All-Sky Solar Radiation	RadAvg	2713	2441	2011	1507	1126	954	1061	1338	1815	2261	2608	2798	
(45)	RadStd	80	81	65	77	61	60	51	60	77	82	94	115	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	-1.87	N/A	N/A	N/A	N/A

Nomenclature: See separate page